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EXPERIMENT 4

Aim: To analyze original and mod APK of using Quark.

Theory:

What is Quark

Quark is a static analysis tool designed to detect malicious activities and vulnerabilities in Android applications by scanning APK files. It operates using a rule-based detection system that identifies suspicious patterns in an app's code without executing it. Unlike dynamic analysis tools, Quark does not require a runtime environment, making it a fast and lightweight option for security researchers and penetration testers. It specializes in uncovering privacy leaks, backdoors, spyware, and privilege escalation risks, making it a valuable asset in Android malware analysis and security auditing.

APK Analysis using Quark

APK analysis using Quark involves scanning an application's code for predefined malicious signatures and behavior patterns. The tool inspects API calls, permissions, and embedded scripts to flag potential security threats. The output provides detailed insights into detected risks, such as unauthorized data collection, suspicious network activity, and privilege abuse. Since it focuses on static code analysis, Quark is best used in combination with other tools like MobSF for dynamic testing and JADX for decompilation to gain a more complete security assessment of an Android application.

Output:

- 1) Quark Tool



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```
(root@kali)-[/home/kali/Downloads]
# quark --help

An Obfuscation-Neglect Android Malware Scoring System

Usage: quark [OPTIONS]

Quark is an Obfuscation-Neglect Android Malware Scoring System

Options:
  -s, --summary TEXT          Show summary report. Optionally specify the
                              name of a rule/label
  -d, --detail TEXT           Show detail report. Optionally specify the
                              name of a rule/label
  -o, --output FILE           Output report in JSON
  -w, --webreport FILE        Generate web report
  -a, --apk FILE              APK file [required]
  -r, --rule PATH             Rules directory [default: /root/.quark-
                              engine/quark-rules/rules]
  -g, --graph [png|json]      Create call graph to call_graph_image
                              directory
  -c, --classification        Show rules classification
  -t, --threshold [100|80|60|40|20]
                              Set the lower limit of the confidence
                              threshold
  -i, --list [all|native|custom]
                              List classes, methods and descriptors
  -p, --permission            List Android permissions
  -l, --label [max|detailed]   Show report based on label of rules
  -C, --comparison            Behaviors comparison based on max confidence
                              of rule labels
  --generate-rule DIRECTORY   Generate rules and output to given directory
  --core-library [androguard|rizin]
                              Specify the core library used to analyze an
                              APK
  --multi-process INTEGER RANGE
                              Allow analyzing APK with N processes, where
                              N doesn't exceeds the number of usable CPUs
                              - 1 to avoid memory exhaustion. [x ≥ 1]
  --version                   Show the version and exit.
  --help                      Show this message and exit.
```

2) APK Analysis

```
(kali@kali)-[/Downloads]
$ quark -s flashlight-1-2-3.apk -s

An Obfuscation-Neglect Android Malware Scoring System

100% [ ] WARNING: Moderate Risk [x] Total Score: 211 211/211 [00:55:00:00, 3.80it/s]

File Name | Rule | Confidence | Score | Weight |
|---|---|---|---|---|
| 00064.json | Monitor incoming call status | 40% | 1 | 0.125 |
| 00211.json | Open an SMS in Webview | 20% | 1 | 0.0025 |
| 00184.json | Set camera preview texture | 40% | 1 | 0.125 |
| 00180.json | Load native libraries(.so) via System.loadLibrary (60% means caught) | 20% | 1 | 0.0025 |
| 00079.json | Hide the current app's icon | 40% | 1 | 0.125 |
| 00136.json | Stop recording | 40% | 1 | 0.125 |
| 00054.json | Install other APKs from file | 40% | 1 | 0.125 |
| 00045.json | Query the name of currently running application | 20% | 1 | 0.0025 |
| 00088.json | Create a secure socket connection to the given host address | 60% | 1 | 0.25 |
| 00123.json | Save the response to JSON after connecting to the remote server | 40% | 1 | 0.125 |
| 00067.json | Check the current network type | 40% | 1 | 0.125 |
| 00097.json | Get the sender address of the SMS and put it into JSON | 40% | 1 | 0.125 |
| 00016.json | Get JSON object prepared and fill in location info | 40% | 1 | 0.125 |
| 00202.json | Make a phone call | 100% | 1 | 1.0 |
| 00157.json | Instantiate new object using reflection, possibly used for dexClassLoader | 100% | 1 | 1.0 |
| 00034.json | Query the current data network type | 100% | 1 | 1.0 |
| 00053.json | Implicit intent(view a web page, make a phone call, etc.) via setData | 100% | 1 | 1.0 |
| 00199.json | Stop recording and release recording resources | 20% | 1 | 0.0025 |
| 00089.json | Connect to a URL and receive input stream from the server | 100% | 1 | 1.0 |
| 00163.json | Perform accessibility service action on accessibility node info | 20% | 1 | 0.0025 |
| 00185.json | Start capturing camera preview frames to the screen | 60% | 1 | 0.25 |
| 00173.json | Get bounds in screen of an AccessibilityNodeInfo and perform action | 40% | 1 | 0.125 |
| 00124.json | Check the current active network type | 40% | 1 | 0.125 |
| 00197.json | Set the audio encoder and initialize the recorder | 20% | 1 | 0.0025 |
| 00062.json | Get the current WiFi MAC address | 40% | 1 | 0.125 |
| 00066.json | Query the ICCID number | 40% | 1 | 0.125 |
| 00110.json | Query the ICCID number | 20% | 1 | 0.0025 |
| 00078.json | Get the network operator name | 100% | 1 | 1.0 |
| 00200.json | Capture the contents of the device screen | 20% | 1 | 0.0025 |
| 00077.json | Read sensitive data(SMS, CALLLOG, etc) | 100% | 1 | 1.0 |
| 00016.json | Get location info of the device and put it to JSON object | 100% | 1 | 1.0
```

3) Permissions



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```
(kali@kali)-[~/Downloads]
$ quark -a flashlight-1-2-3.apk -p

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com.google.android.c2dm.permission.RECEIVE
android.permission.WAKE_LOCK
com.google.android.finsky.permission.BIND_GET_INSTALL_REFERRER_SERVICE
com.surpax.ledflashlight.panel.permission.C2D_MESSAGE
android.permission.ACCESS_NETWORK_STATE
android.permission.FLASHLIGHT
com.surpax.ledflashlight.panel.permission.FRAMEWORK_SECURITY
android.permission.INTERNET
```

4) MOD APK Analysis

```
(root@kali)-[/home/kali/Downloads]
$ quark -a Flashlight\ Plus\ v2.8.3\ \Plus\).apk -s

An Obfuscation-Neglect Android Malware Scoring System

Requested API level 33 is larger than maximum we have, returning API level 28 instead.
100%
[!] WARNING: Moderate Risk
[*] Total Score: 211

+-----+-----+-----+-----+-----+
| Filename | Rule | Confidence | Score | Weight |
+-----+-----+-----+-----+-----+
| 00064.json | Monitor incoming call status | 40% | 1 | 0.125 |
| 00211.json | Open an URL in Webview | 20% | 1 | 0.0625 |
| 00184.json | Set camera preview texture | 60% | 1 | 0.25 |
| 00180.json | Load native libraries(.so) via System.loadLibrary (60% means caught) | 60% | 1 | 0.25 |
| 00079.json | Hide the current app's icon | 100% | 1 | 1.0 |
| 00136.json | Stop recording | 20% | 1 | 0.0625 |
| 00054.json | Install other APKs from file | 40% | 1 | 0.125 |
| 00045.json | Query the name of currently running application | 20% | 1 | 0.0625 |
| 00088.json | Create a secure socket connection to the given host address | 20% | 1 | 0.0625 |
| 00123.json | Save the response to JSON after connecting to the remote server | 40% | 1 | 0.125 |
| 00097.json | Check the current network type | 100% | 1 | 1.0 |
| 00097.json | Get the sender address of the SMS and put it into JSON | 40% | 1 | 0.125 |
| 00018.json | Get JSON object prepared and fill in location info | 40% | 1 | 0.125 |
| 00202.json | Make a phone call | 100% | 1 | 1.0 |
| 00157.json | Instantiate new object using reflection, possibly used for dexClassLoader | 100% | 1 | 1.0 |
```

Conclusion:

Thus we successfully analyzed original and mod APK of using Quark. Quark effectively detects static threats, combining it with dynamic analysis and reverse engineering tools can provide a more comprehensive security assessment, making it a crucial step in evaluating APK authenticity and safety.